

EE7500: Advanced Electromagnetics — Homework Assignment 1

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Total: 100 points

Academic Integrity:

- This homework must be completed and submitted individually.
- If you discuss the general approach with classmates, [acknowledge them by name](#) in your submission (no penalty for doing so).
- If you use AI tools (ChatGPT, Claude, etc.), clearly state which problems, how you used them, and the specific prompts. *If you use machine learning tools as a black box without building your own understanding, it is only the machines that will learn—not you.*
- Submit: PDF with all solutions/plots, and MATLAB .m files for computational problems as one consolidated zip file. The matlab files should have comments and named like so: 'questionXX.m' and the PDF file name must be your roll number.
- Your main PDF should start with a self-certification of academic integrity like so: "I hereby certify that this document and associated code has been generated by me with the highest academic integrity – your name"

1. Transmission Line Analysis

A lossless transmission line with $Z_0 = 50 \Omega$ at $f = 1 \text{ GHz}$ is terminated with $Z_L = 75 + j50 \Omega$. The line has length $\ell = 0.15\lambda$.

- (3 points) Calculate the load reflection coefficient Γ_L in both rectangular and polar forms, and the VSWR.
- (3 points) Find the input impedance Z_{in} .
- (2 points) If $P_{inc} = 10 \text{ W}$, find the power delivered to the load and the power reflected.
- (2 points) Sketch the voltage magnitude $|V(d)|$ along the line as a function of distance d from the load, indicating the locations of the first maximum and first minimum.

2. Quarter-Wave Transformer Design

Design a quarter-wave matching network: $Z_L = 100 \Omega$ to $Z_0 = 50 \Omega$ at $f_0 = 2.4 \text{ GHz}$ on a substrate with $\epsilon_r = 4.4$ (FR-4).

- (2 points) Calculate the characteristic impedance Z_{QW} of the quarter-wave section.
- (2 points) Calculate the physical length of the matching section in millimeters.
- (c) (2 points) Estimate the fractional bandwidth over which $|\Gamma_{in}| < 0.1$.
- (2 points) Draw a labeled schematic showing the source line (Z_0), the quarter-wave matching section (indicating its impedance and physical length), and the load (Z_L).

3. Skin Depth and Shielding Effectiveness

A two-layer planar shield is illuminated by a normally-incident plane wave at $f = 200 \text{ MHz}$. Layer 1 (outer): aluminum, $t_1 = 100 \mu\text{m}$, $\sigma_{Al} = 3.5 \times 10^7 \text{ S/m}$. Layer 2 (inner): copper, $t_2 = 50 \mu\text{m}$, $\sigma_{Cu} = 5.8 \times 10^7 \text{ S/m}$. Both have $\mu_r = 1$; negligible air gap.

- (3 points) Calculate the skin depth $\delta_s = 1/\sqrt{\pi f \mu_0 \sigma}$ for Al and Cu at 200 MHz. Express each layer thickness in units of skin depth, then compute the absorption shielding effectiveness $SE \approx 8.686 (t/\delta_s) \text{ dB}$ for each layer and the total SE.

- (b) (3 points) Compare with a single 150 μm aluminum layer (same total thickness). Which configuration provides better shielding? Explain physically why.
- (c) (2 points) At $f = 5\text{ GHz}$, a plane wave with surface magnetic field $H_0 = 1\text{ A/m}$ is incident on a thick copper slab. Using the impedance boundary condition $Z_s = (1 + j)/(\sigma\delta_s)$, compute $\vec{E}_{tan}$ at the surface.
- (d) (2 points) The absorption-only SE formula used above ignores re-reflections at layer boundaries. Under what condition on t/δ_s is this approximation justified? Is it valid for the shield in this problem?

Ignore reflection losses at boundaries for the SE calculations.

4. Polarization and Wave Superposition

A plane wave at $f = 10\text{ GHz}$ propagates in the $+z$ direction in free space. Its phasor electric field is $\vec{E} = (a_x \hat{x} + a_y e^{j\delta} \hat{y}) e^{-jk_0 z}\text{ V/m}$, with $e^{j\omega t}$ time convention.

- (a) (3 points) For $a_x = 3$, $a_y = 4$, $\delta = 0$: identify the polarization type. Find the tilt angle ψ of the electric field vector relative to the x -axis and compute the time-averaged power density $|\vec{S}_{av}|$.
- (b) (3 points) For $a_x = 3$, $a_y = 4$, $\delta = -\pi/2$: write the time-domain electric field $\vec{E}(z=0, t)$ and evaluate it at $t = 0, T/4, T/2, 3T/4$. Sketch the locus of the $\vec{E}$ -field tip in the x - y plane. Identify the polarization type, the sense of rotation (right-hand or left-hand), and compute the axial ratio.
- (c) (3 points) Two co-polarized plane waves of the same frequency arrive at a receiver:

$$\begin{aligned}\vec{E}_1 &= 2 \hat{x} e^{-jk_0 z} \text{ V/m (direct path),} \\ \vec{E}_2 &= 3 \hat{x} e^{-j(k_0 z + \pi/3)} \text{ V/m (reflected path with additional phase } \pi/3\text{).}\end{aligned}$$

Compute the time-averaged Poynting vector magnitude for each wave individually ($|\vec{S}_1|$, $|\vec{S}_2|$) and for the total field $\vec{E} = \vec{E}_1 + \vec{E}_2$. Verify that $|\vec{S}_{total}| \neq |\vec{S}_1| + |\vec{S}_2|$ and explain physically why superposition does not apply to the Poynting vector.

- (d) (3 points) The reflected wave now has a different polarization due to ground reflection:

$$\begin{aligned}\vec{E}_1 &= 2 \hat{x} e^{-jk_0 z} \text{ V/m (horizontally polarized, direct path),} \\ \vec{E}_2 &= 1.5 \hat{y} e^{-j(k_0 z + \pi/4)} \text{ V/m (vertically polarized, reflected path).}\end{aligned}$$

What is the polarization of the total field $\vec{E}_1 + \vec{E}_2$? Is it linear, circular, or elliptical? Determine the sense of rotation.

5. Vector Potential from a Current Distribution

A line current along the z -axis: $\vec{J}(\vec{r}') = I_0 e^{-|z'|/a} \delta(x') \delta(y') \hat{z}$, where I_0 has units of A and a is a length.

- (a) (3 points) Using the 3D free-space Green's function $G(\vec{r}, \vec{r}') = \frac{e^{-jk|\vec{r}-\vec{r}'|}}{4\pi|\vec{r}-\vec{r}'|}$, write the integral for the vector potential $\vec{A}(\vec{r}) = \mu \int_V \vec{J}(\vec{r}') G(\vec{r}, \vec{r}') dV'$.
- (b) (2 points) Simplify by performing the x' and y' integrations. State the remaining integral and its limits.
- (c) (3 points) Compute $\nabla' \cdot \vec{J}$ and show it is *not* zero. Using the continuity equation $\nabla \cdot \vec{J} = -j\omega\rho$, what does this imply about the charge density and scalar potential ϕ ?
- (d) (2 points) Explain physically why $\vec{A}$ must point in the $\hat{z}$ direction everywhere for this source.

You do not need to evaluate the integral in (b).

6. Verification of 1D Green's Function

The 1D Green's function $g(x, x') = \frac{-j}{2k_0} e^{-jk_0|x-x'|}$ satisfies $\frac{d^2g}{dx^2} + k_0^2g = -\delta(x - x')$.

- (a) (3 points) For $x > x'$, write g without the absolute value. Compute dg/dx and d^2g/dx^2 , then substitute into $d^2g/dx^2 + k_0^2g$ and show it equals zero.
- (b) (2 points) Repeat the above for $x < x'$.
- (c) (2 points) Compute the jump in dg/dx at $x = x'$: $\left. \frac{dg}{dx} \right|_{x'+} - \left. \frac{dg}{dx} \right|_{x'-}$. Show it equals -1 .
- (d) (3 points) Integrate the defining equation over $[x' - \epsilon, x' + \epsilon]$ and take $\epsilon \rightarrow 0$. Show that the result is consistent with part (c).

7. 1D Green's Function — MATLAB Symbolic Toolbox

Use MATLAB's Symbolic Math Toolbox to verify properties of the 1D Green's function.

Reference: <https://www.mathworks.com/help/symbolic/>

- (a) (2 points) **Symbolic definition.** Define $g_1(x)$ for $x > x'$ and $g_2(x)$ for $x < x'$ as symbolic expressions (with x, x', k_0 as symbolic variables).
- (b) (3 points) **Symbolic verification of Helmholtz equation.** For each region: compute dg/dx , d^2g/dx^2 , then $d^2g/dx^2 + k_0^2g$. Use `simplify` to verify the result is zero. Display expressions at each step.
- (c) (2 points) **Derivative discontinuity.** Symbolically evaluate dg_1/dx at $x = x'$ (from right) and dg_2/dx at $x = x'$ (from left). Compute the jump and verify it equals -1 .
- (d) (3 points) **Numerical plotting.** Convert to numeric using `matlabFunction`. For $k_0 = 2\pi, x' = 0$, plot $\text{Re}\{g\}$ and $\text{Im}\{g\}$ as two subplots over $x \in [-3, 3]$. Verify $g(1, 0) = g(0, 1)$ numerically (symmetry).

Deliverables: MATLAB .m file with comments, intermediate symbolic outputs printed to the command window, plots with labeled axes, and a 2–3 sentence summary of what the symbolic verification confirms about the Green's function.

8. Sommerfeld Radiation Boundary Condition

The general homogeneous solution for the 2D Green's function is $g(r) = a H_0^{(1)}(kr) + b H_0^{(2)}(kr)$.

The Sommerfeld radiation boundary condition (for $e^{j\omega t}$ convention) is: $\lim_{r \rightarrow \infty} \sqrt{r} \left(\frac{\partial g}{\partial r} + jkg \right) = 0$.

Asymptotic forms: $H_0^{(1)}(kr) \sim \sqrt{\frac{2}{\pi kr}} e^{j(kr - \pi/4)}$; $H_0^{(2)}(kr) \sim \sqrt{\frac{2}{\pi kr}} e^{-j(kr - \pi/4)}$.

- (a) (2 points) For $H_0^{(2)}(kr)$: compute $\frac{\partial}{\partial r} H_0^{(2)}(kr)$ using the asymptotic form (leading order), substitute into the radiation condition, and verify it is satisfied.
- (b) (2 points) Repeat for $H_0^{(1)}(kr)$ and show it does *not* satisfy the condition.
- (c) (4 points) The DLMF (<https://dlmf.nist.gov/10.17>) gives higher-order asymptotic expansions for the Hankel functions (which equations are relevant?). Write out the *two-term* expansion of $H_0^{(2)}(kr)$ and show that it also satisfies the Sommerfeld radiation condition.
- (d) (2 points) Why are Hankel functions preferred over Bessel functions J_0, Y_0 for radiation problems?

Hint for (a)–(b): Use the product rule; keep only the leading-order term in $1/r$.

9. Dimensional Comparison of Green's Functions

The free-space Green's functions are:

$$1\text{D: } g_{1D} = \frac{-j}{2k} e^{-jk|x-x'|}, \quad 2\text{D: } g_{2D} = \frac{-j}{4} H_0^{(2)}(k|\vec{r} - \vec{r}'|), \quad 3\text{D: } g_{3D} = \frac{e^{-jk|\vec{r} - \vec{r}'|}}{4\pi|\vec{r} - \vec{r}'|}$$

In the far field, $|g_{2D}| \sim 1/\sqrt{2\pi kr}$ for $r \gg \lambda$.

- (a) (2 points) Extract the amplitude decay factor $|g|$ vs. distance for each case and state the decay law (constant, $1/\sqrt{r}$, $1/r$).
- (b) (3 points) Using energy conservation, derive the decay rates from first principles:
 - 1D: power flows along one direction only.
 - 2D: power spreads over a cylindrical surface of circumference $2\pi r$.
 - 3D: power spreads over a sphere of area $4\pi r^2$.

Since field amplitude $\propto \sqrt{\text{intensity}}$, verify consistency with part (a).

- (c) (3 points) A finite line source of length L radiates in 3D space. In the near field ($r \ll L$) it looks like an infinite line source; in the far field ($r \gg L$) it looks like a point source. Which Green's function describes each regime? Estimate the transition distance.
- (d) (2 points) Explain in 2–3 sentences:
 - Why the 1D Green's function has no amplitude decay.
 - Why all real (finite-extent) antennas eventually exhibit 3D $1/r$ decay in the far field.

10. 1D Green's Function in a Layered Medium

Consider the 1D Helmholtz equation with a piecewise-constant wavenumber modeling an interface between two non-magnetic ($\mu_r = 1$) dielectric half-spaces:

$$\frac{d^2 g}{dx^2} + k^2(x) g = -\delta(x - x'), \quad k(x) = \begin{cases} k_1 = \omega \sqrt{\mu_0 \varepsilon_0 \varepsilon_{r1}}, & x < 0 \\ k_2 = \omega \sqrt{\mu_0 \varepsilon_0 \varepsilon_{r2}}, & x > 0 \end{cases}$$

A point source is located at $x = x' > 0$ (inside medium 2). Both g and dg/dx are continuous at the interface $x = 0$ (TE polarization, same μ).

- (a) (3 points) Write the general solution for $g(x)$ in each of the three regions: $x < 0$, $0 < x < x'$, and $x > x'$. Apply the outgoing-wave (radiation) condition at $x \rightarrow \pm\infty$ to discard the appropriate terms. How many unknown coefficients remain?
- (b) (3 points) Write the four boundary conditions (two at $x = 0$, two at $x = x'$) and solve for the coefficients. Express your answer in terms of k_1 , k_2 , x' , and the reflection coefficient $\Gamma = (k_2 - k_1)/(k_1 + k_2)$.
- (c) (2 points) Verify that when $\varepsilon_{r1} = \varepsilon_{r2}$ (homogeneous medium, $\Gamma = 0$), your solution reduces to the free-space Green's function $g = \frac{-j}{2k} e^{-jk|x-x'|}$.
- (d) (2 points) For $\varepsilon_{r1} = 1$ and $\varepsilon_{r2} = 4$: compute Γ and describe qualitatively how $|g(x)|$ differs from the homogeneous case. In which region does a partial standing wave appear, and why? What happens to the amplitude transmitted into medium 1?